

Image Processing – HW2

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Please refer to the README.md first, the following only explain the code logic and the reason for the result.

Q1:

Steps:

1. Make it binary
2. Closing operation to get the repaired image
3. Apply Dilation to that repaired image then subtract afterwards to get outer boundary

Reason:

It's just basic OpenCV function call with basic morphological operations, all as expected. For images, please see README.md to try to locate.

Q2:

Steps:

1. Stretch the local min, max to 0 and 255 linearly

Reason:

```
Original min: 0, Original max: 255  
---Stretch to---  
Global min: 0, Global max: 255
```

As you can see, there's no difference between the contrast stretched version and the original one because the local min and max for the original image is 0 and 255. Applying linear contrast stretching will do absolutely nothing.

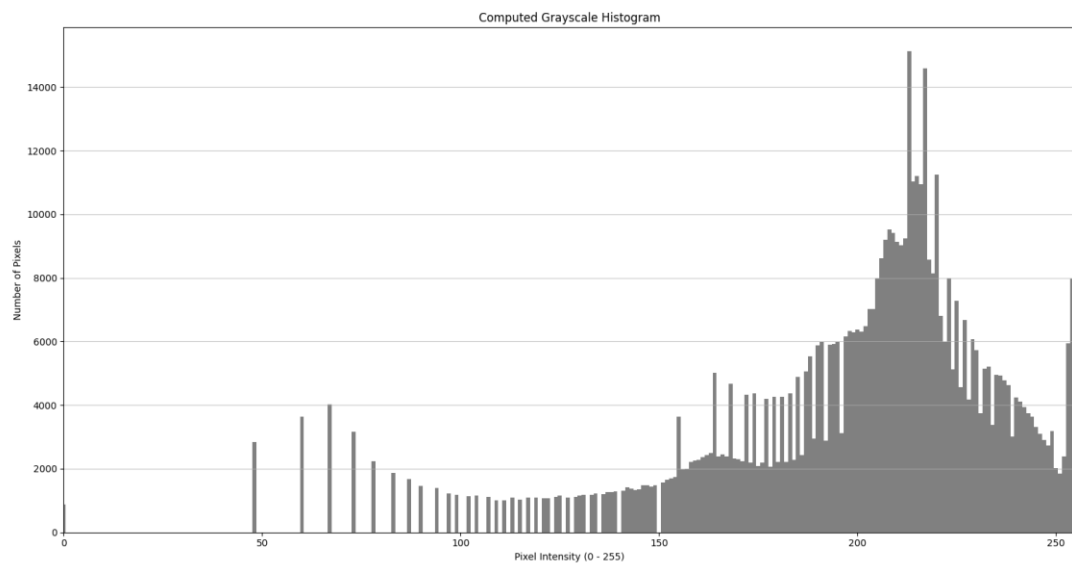
Q3:

Steps:

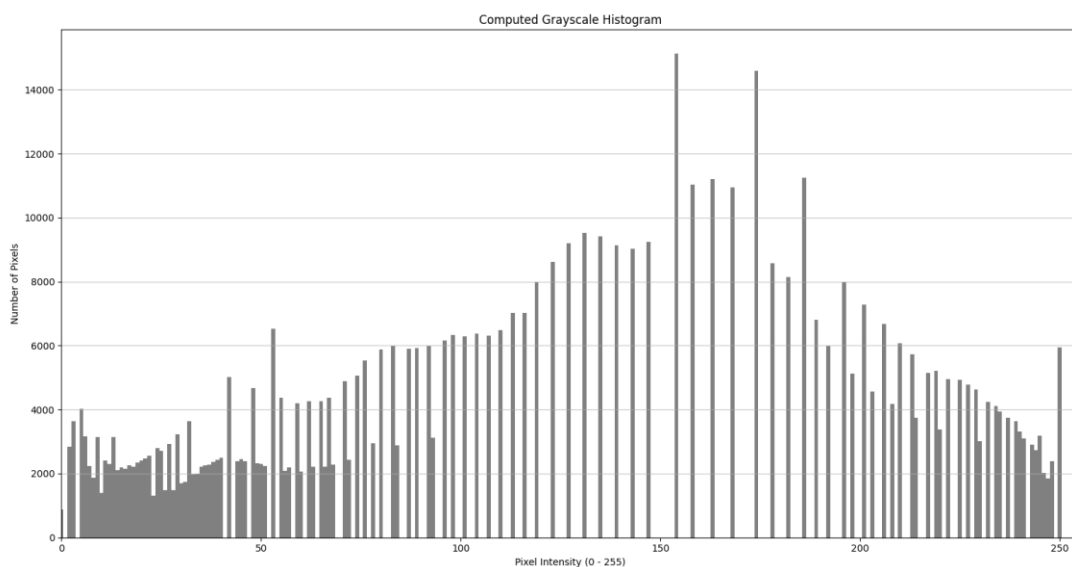
1. Generate the histogram from input images
2. Sum the probability to get CDF
3. Creating an intensity lookup table
4. Apply mapping (histogram specification)

Reason:

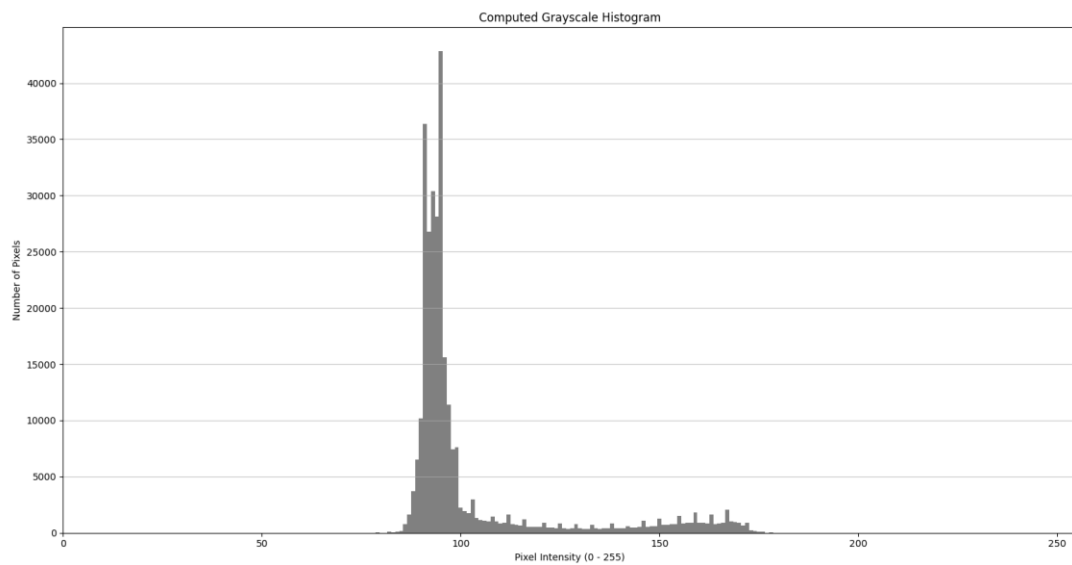
For aerialview-washedout.tif, it's working very well because



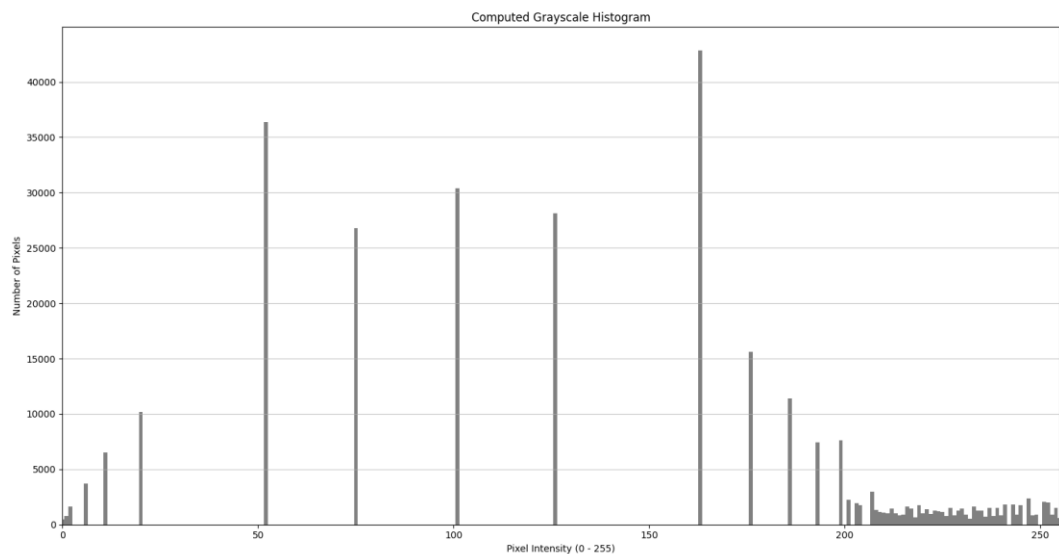
As you can see, the overall intensities are pretty spread out. Therefore, it can be mapped effectively without a particular peak.



For einstein-low-contrast.tif, it's the opposite. Take a look at the original histogram:



Almost all pixels are in 90 ~ 100. When we GHE this, the image will be overly bright just to match this heavily skewed case.



As you can see, lots of pixels are around 200 ~ 250 just to satisfy the CDF.

Q4:

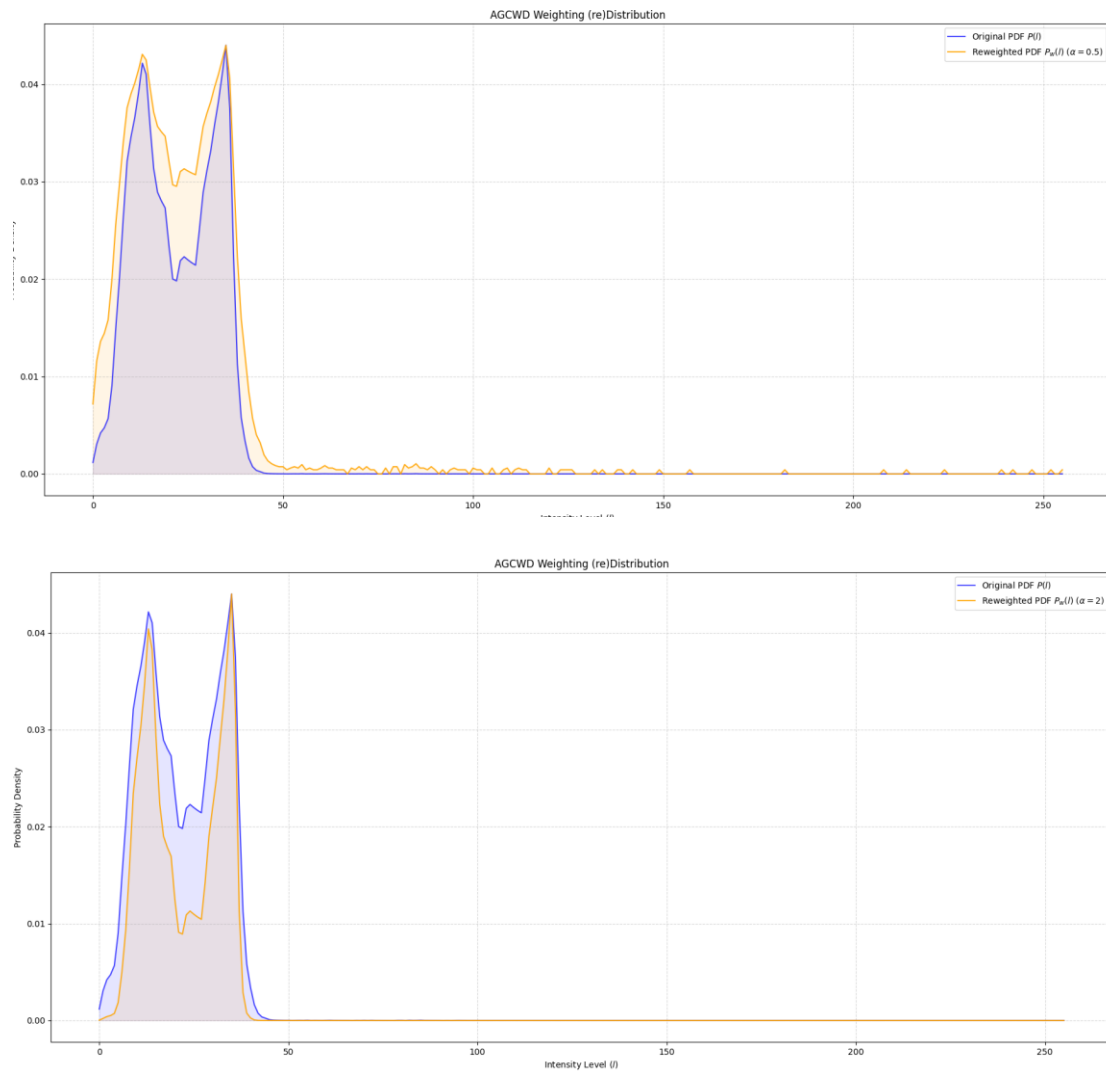
Steps:

1. Generate the histogram from input images

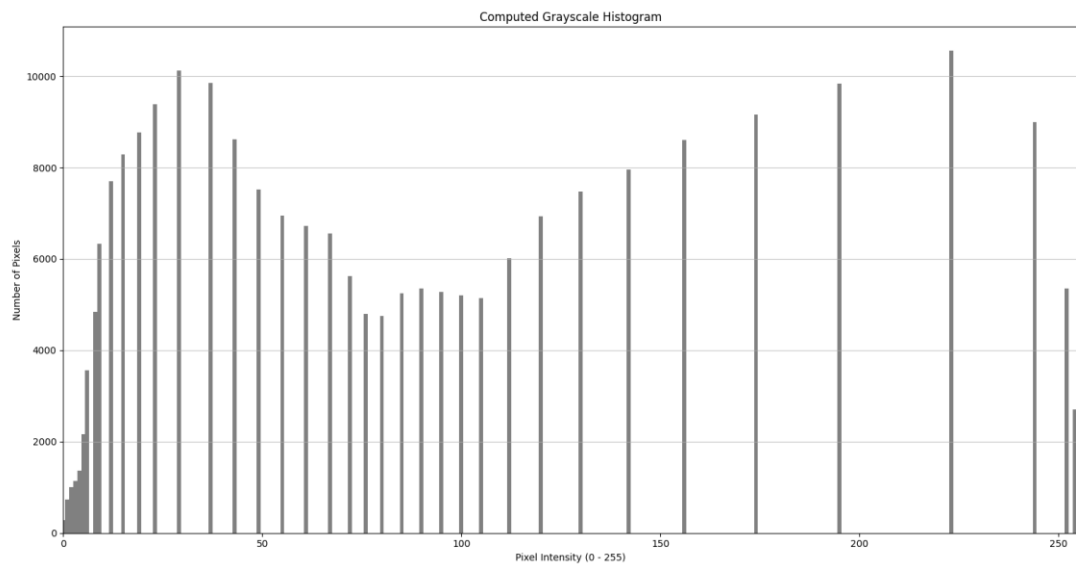
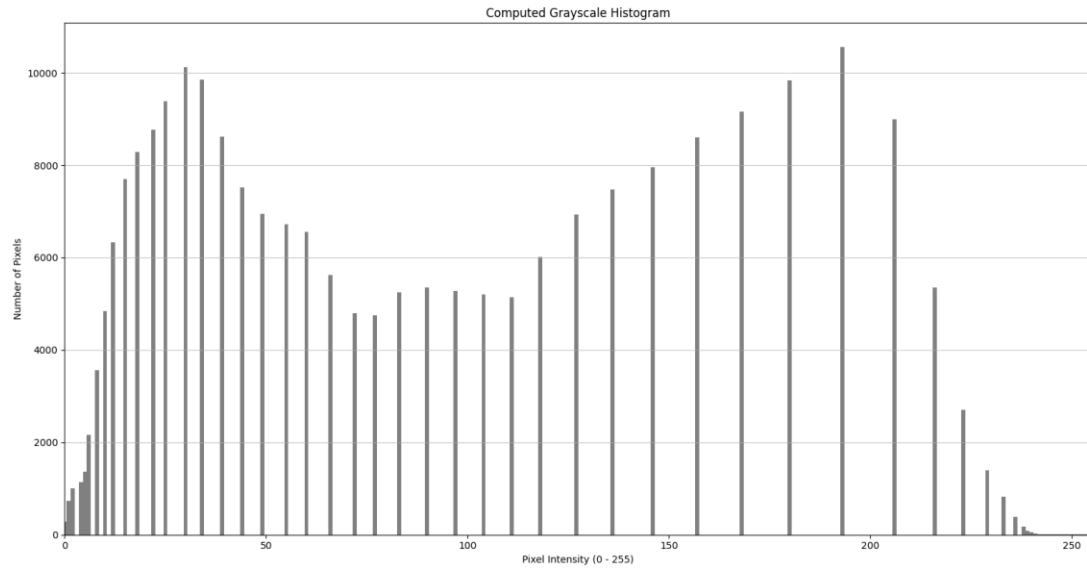
2. Re-weight using alpha
3. Calculate the CPF_w
4. Apply Gamma correction for each intensity

Reason:

It seems weird that the 2 images with different alpha (0.5 and 2) looks almost the same, but here's the breakdown:



As you can see, the re-weighted PDF only makes a difference in intensities around 25 ~ 30, which is not that obvious for human eyes to differentiate. Look at the AGCWD images' histogram too:



Not really much of a difference.

Q5:

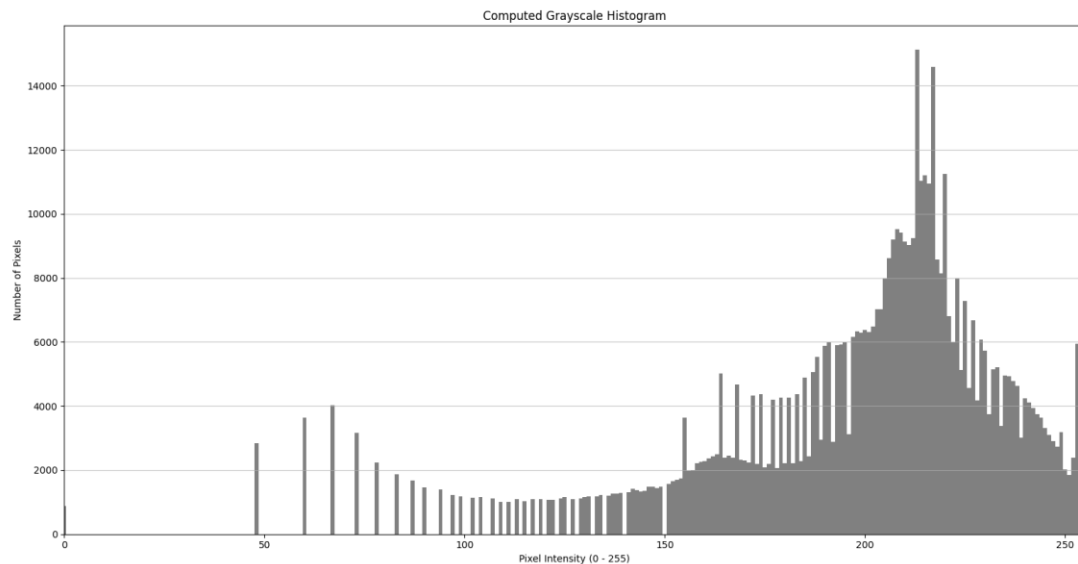
Steps:

1. Get mean, median intensity
2. Just like Q3 but we set the start and end intensity zone
3. Since the 2 partitions won't overlap, we can just get 2 images using 2 HE and add them together

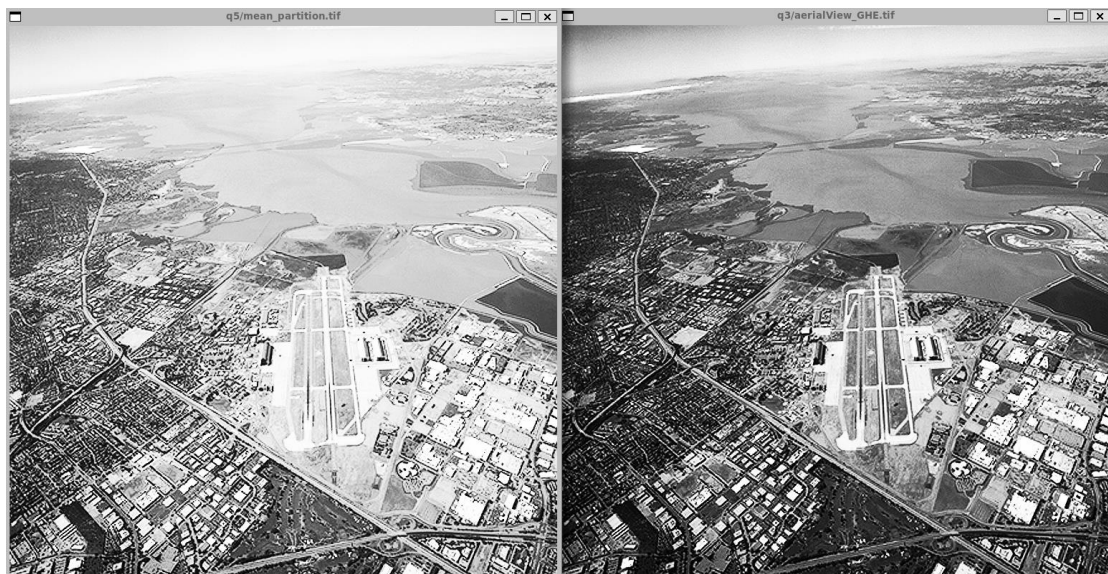
Reason:

Mean: 197
Median: 208

Since the mean and median are quite similar, we can discuss them together (use Median as example). The original image's histogram looks like:



By comparing the partition HE with Q3's GHE:



We can see that the overall color is brighter than the GHE. Because the original histogram is centered on the right, which gets pulled leftward after GHE, and partition HE will make sure that intensities over a threshold (In this median's case: 197) won't get pulled left to it (no less than 197).

Q6:

Steps:

1. Construct 2D Histogram using np's advanced indexing
 - Especially in applying kernels to the image, instead of padding, we set boundary mask so that center_vals and neighbor_vals are talking to each other conveniently.
2. Normalization (in 2D)
3. Compress the 2D normalized histogram to a 1D CDF array
4. Creating an intensity lookup table
5. Apply mapping (histogram specification)

Reason:

It's working pretty well.



It looks like spatial information is preserved because the components of the image are obviously more natural than the one in question 3's GHE.

